Design Decisions & User Journeys

This document explains **how the platform is used** (the three kinds of user and what each does, start to finish) and **why it is built the way it is** (the design decisions behind the parts that aren't obvious). It also covers two features the other handoff documents don't: the **Alignment** and **Notifications** tabs.

Unfamiliar terms (Lambda, ARN, Bedrock, BN, …) are defined in the glossary in the README. Companion documents: 01 · Project Audit, 02 · Deploy Runbook, 03 · Engine Comparison, 04 · Monitoring & Security.

Part A — The three user journeys

The platform serves **three kinds of user** (each login is exactly one kind). They see different pages and can take different actions; a shared, public **RAP Index** and **Legal Cases** library sit underneath all three.

1. Business / company

A company that publishes a Reconciliation Action Plan (RAP) and reports its Indigenous-supplier spend.

1. **Report supplier spend** — on their questionnaire they add itemized lines, each naming an Indigenous supplier, an amount, and a flow type (procurement or capital). New lines start as **"pending"** — unconfirmed until the named supplier responds.
2. **Watch coverage** — a read-only view shows what share of their reported spend has been *confirmed* by suppliers. It is framed as "a data view, not a rating."
3. **Track commitments** — they add and update their own RAP commitments, see **overdue / at-risk** alerts, and see **suggested Indigenous suppliers** for each procurement commitment (the Alignment feature, Part C).
4. **Claim their organization** — they enter their federal **Business Number (BN)** to claim the org. Claiming is what authorizes them to post progress against their published RAP.
5. **Upload & track their RAP** — they upload their RAP PDF (it goes to the Institute for AI-extraction and review), then record progress against the published, grounded commitments. The extracted text itself is read-only to them; they add progress on top. They can optionally opt their RAP onto the public Index.

2. Supplier (Indigenous business)

An Indigenous-owned business that confirms the spend companies report against it and showcases its record.

1. **Confirm inbox** — they see every line a company reported naming them, and respond: **Confirm / Dispute / Correct** (correction carries a corrected amount). The rule is *"silence is never confirmed"* — nothing counts as confirmed unless the supplier acts.
2. **My Record (data sovereignty)** — confirmed revenue accrues here; they can **export** their data (JSON) and **withdraw** confirmations at any time. This is a deliberate OCAP® gesture — the supplier owns and controls their data.

3. **Profile / showcase** — they edit a public profile (sector, region, website, blurb), toggle it public, and **claim certifications** (CCIB, ISC Indigenous Business Directory, a Nation, or a regional body) by linking an external reference — which stays *pending* until the Institute reviews it.
4. **Public showcase page** — a shareable public page shows their verification tier, confirmed track record, and spend breakdown.

3. Indigenomics Institute (staff)

The steward that reviews extractions, verifies suppliers, monitors the network, and brokers matches.

1. **RAP Index (landing)** — the network-wide dashboard: key takeaways, KPIs, status and confirmation-integrity charts, deadline/risk tables, sector/type breakdowns, and a full filterable commitment list with CSV export.
2. **Organizations & suppliers** — per-organization RAP scorecards, and a searchable directory of verified suppliers.
3. **Extract & review** — upload RAP PDFs and run AI extraction. Clean, high-confidence extractions **auto-publish**; anything the AI was unsure about is **flagged into a review queue** for a human to check against the source PDF and publish (guided by the reviewer guide at `/extract/guide`).
4. **Verify suppliers** — approve or reject supplier certification claims ("we verify the link; we don't re-certify").
5. **Alignment** — broker introductions between company procurement commitments and fitting verified suppliers (Part C).
6. **Coverage & integrity analytics** — confirmed-vs-reported spend, spend by ownership-certification tier, and integrity signals (e.g. certified-but-no-activity, self-declared-with-activity).
7. **Notifications** — generate/review the weekly overdue-and-at-risk milestone digest (Part C).

   Small known rough edges (worth noting for a future team): an internal `personaHome()` helper is defined but no longer used for routing (the real post-login path is `/home`, which sends the Institute to the RAP Index); the public landing page links to the RAP Index, but a logged-out visitor who clicks it is sent to the login page first; and the coverage/analytics view is readable by any signed-in user, not only the Institute. None affects data integrity; all are easy tidy-ups.

Part B — Design decisions (the "why")

B.1 Business Number (BN) as organization identity — resolved in the review queue

Decision. A published organization is keyed on its **federal Business Number**, resolved during review (with a real Business-Number checksum pre-check and a pre-filled federal registry lookup link), **not** on the organization name the AI read from the document. *(The automated ISED registry verification is scaffolded but **not yet activated** — the default provider is a stub, so today an entity the registry can't confirm resolves as self-asserted; the checksum pre-check does run.)*

Why. *A name is not an identity.* The same brand can hide several legal entities, and one entity can be written several ways: - **False merge** — a RAP says "Enbridge," but Corporations Canada lists *three* distinct entities (Enbridge Inc., Enbridge Pipelines Inc., Enbridge Frontier Inc.), each with its own BN. A name key could attach commitments to the wrong legal entity. - **False split** — "Enbridge Inc." vs "Enbridge Gas Inc." fragments one entity into two.

Only the BN disambiguates. Resolving it **at review** (before a RAP becomes canonical) means identity is pinned by a human at exactly the moment the document enters the record — and the same BN is what later authorizes the company to post progress. Entities the federal registry can't confirm (e.g. provincial-only) publish as **self-asserted**, clearly badged, rather than being blocked.

B.2 Human-in-the-loop review gate — not full auto-publish

Decision. Clean, high-confidence extractions publish automatically; anything the AI flags goes to a **human review queue** before publishing.

Why. AI extraction is strong but not perfect, and this is a provenance-first, citation-anchored platform where a wrong commitment attributed to a real company matters. Critically, our own study (see 03) found that using a **second AI to "check" the first does not work on this data** (the two AI judges agreed no better than chance, κ=0). So a human — not an automated AI check — is the correct arbiter. Grounding (B.4) makes this fast: each flagged field shows the exact quote and page to verify against.

B.3 Two-to-three extraction engines — residency, provenance, and fallback

Decision. The platform can read documents through more than one engine, chosen along **two independent switches**: *which AI extraction path* (`EXTRACTION_IMPL` = BDA / Bedrock / mock) and *which document reader* (`DOC_LOADER` = Textract OCR / text-layer). This yields the three real engines compared in 03: **BDA**, **Bedrock + Textract-LAYOUT**, and **Bedrock + text-layer**.

Why not just one? Three different pressures pull in different directions: - **Data residency.** In the current sandbox, AWS Textract is blocked for automated (Lambda) use by the parent organization's policy, so the **text-layer** reader — which needs no AWS OCR service — is the only fully in-country automated reader. It exists precisely so the Canadian stage can run without the blocked service. *(This block disappears in a client-owned account — see 01 §4.3.)* - **Provenance / grounding fidelity.** The verbatim-quote readers (Textract-LAYOUT and text-layer) ground far better than BDA, which grounds by confidence and infers page numbers — a real difference for a citation-anchored product. - **Robustness & cost/speed.** The engines fail on *different* documents and trade cost against speed differently, so having more than one is a genuine fallback, not redundancy.

The measured recommendation (Bedrock + Textract-LAYOUT primary, text-layer as the cheapest/most in-country fallback, BDA only where speed beats provenance) is in 03.

B.4 Grounding & provenance — verbatim quotes, and three separate claims

Decision. Every extracted field carries the **verbatim quote and page** it came from, and the system keeps three *distinct* kinds of claim separate and never conflates them:

1. **Extracted-by** — who/what produced the field (the AI pipeline + the human reviewer).
2. **Progress-reported-by** — the company's self-reported progress, an *append-only* layer on top of the immutable extraction.
3. **Confirmed-by** — independent confirmation (a supplier confirming reported spend).

Why. Trust in this data depends on being able to trace every value to its source and to know *who* is asserting *what*. A company's optimistic progress note must never look like an independently verified fact, and the grounded extraction must never be silently rewritten.

B.5 Evidence precedence — confirmed > research > self-reported

Decision. Everything shown for an organization resolves to one of three tiers, which fixes both how it's displayed and whether it counts toward rankings:

Tier	Source	Ranks?
1 · Confirmed	a supplier confirmed the reported procurement spend	**yes**
2 · Research	the commitment as recorded (seeded or company-created)	**yes**
3 · Self-reported	a company's own uploaded-RAP progress (opted-in orgs only, shown as separate rows)	**no**

Why. Self-report should be *visible* but must never move the ranking — only independent confirmation elevates a commitment. This keeps the public Index honest: a company cannot climb the leaderboard by marking its own homework. Surfacing an uploaded RAP on the public Index is **opt-in** by the company.

B.6 Hybrid ownership — staff and companies both upload; only a claimed company posts progress

Decision. Both Institute staff and companies can upload RAPs; but only a company that has **claimed its BN** can post progress, and its progress is append-only and never edits the grounded extraction.

Why. It shares the workload (the Institute isn't the sole data-entry bottleneck) while keeping a clear trust boundary: the objective, grounded extraction stays under stewardship, and the subjective progress layer is clearly owned by, and attributed to, the company.

B.7 Canadian residency split — data in Canada, AI inference is not

Decision. Platform data can rest in `ca-central-1` (Canada), but the AI-inference step routes to a US/global region.

Why. It's an AWS reality, not a design preference: Amazon's Bedrock models are **not hosted in Canada**, so "in Canada" here means *data-at-rest and document-reading in Canada; the AI model call still leaves Canada.* This is the same for every engine. Detail in 01 §4.2 and 03.

B.8 Serverless single-table database + mock-first development

Decision. The app runs serverless (Lambda + CloudFront + DynamoDB, one table per domain) and defaults to an **in-memory mock** backend unless real-backend flags are set.

Why. Serverless keeps the non-AI running cost near zero (~$1/month — see 01 §3); mock-first lets the whole app run locally with no AWS account or cloud calls, which made development and demos fast and cheap and keeps onboarding simple.

B.9 Edge protection & self-monitoring

Decision. A Web Application Firewall (WAF) sits in front of the site, shipped **count-first** (watching, not yet blocking), and the extraction pipeline monitors its own health with alerts.

Why. Both were added deliberately and conservatively — the WAF counts before it blocks so real users aren't caught by a false positive, and monitoring exists because a silent extraction failure once went unnoticed. Full detail in 04.

Part C — Alignment & Notifications (features not covered elsewhere)

Both are **Institute-facing**, both are real and working, and both carry caveats worth knowing.

C.1 Alignment — the supplier-matchmaking radar

What it does. Alignment matches a company's RAP **procurement commitment** to **verified Indigenous suppliers** that fit it, so the Institute can broker introductions. Each card is one company commitment with its top-ranked suppliers, each showing a *fit %* and a one-line rationale. The same engine also powers a per-company "suppliers that fit this commitment" panel on the company side.

How it works. When a commitment changes, a database trigger recomputes matches in near-real-time. The fit score is a **deterministic weighted sum** — sector match (heaviest), lexical relevance (a genuine BM25 keyword-overlap score), identity-tier trust (Nation > CCIB > self-declared), and ownership share — keeping the strongest few per commitment.

> **Caveats.** - **"AI-matched" overstates the default.** Optional AI (semantic embeddings + an LLM-written rationale) only activates when real AI providers are configured. In the **default deployed configuration these are stubbed**, so the score is fully deterministic (sector + keyword overlap + tier + ownership) and the rationale is a template sentence built from real facts — not live AI. - **Procurement commitments only** — capital/equity commitments are out of scope. - **Verified suppliers only** — self-declared suppliers are filtered out of matches. - **Needs real supplier data to be credible** — until a real set of Indigenous suppliers is seeded, matches are fixture-quality, and the radar can legitimately show as empty. - **Demo-scale** — the Institute radar ranks across a capped set (top ~100 commitments); weights are hand-tuned with no feedback/learning loop; unknown ownership is assumed at bare majority (51%).

Future enhancement — an evidence-based "track record" signal. Today the fit score is *attribute-based* only: it scores whether a supplier *looks* like a fit (sector, keywords, identity tier, ownership), not whether it has *actually delivered*. The platform already captures the raw material to do better — the **confirmed procurement lines** (a company reports spend naming a supplier, the supplier confirms it) are a real collaboration history. A "track record" term could raise suppliers with confirmed deliveries, weighted toward the same sector and comparable companies, making matches evidence-based rather than attribute-based. It should **augment, not dominate**, the score, for three reasons: (a) **cold start / incumbent bias** — new suppliers have no history, so over-weighting it entrenches "rich-get-richer" and works against surfacing new Indigenous businesses (the platform's goal); (b) **same-pair vs cross-company** — a prior deal between *this* supplier and *this* company is a near-redundant match, whereas deliveries to *other* companies are the real capability proof, and the two should be weighted differently; (c) **data sparsity** — at demo scale there is little confirmed data yet, so the signal stays thin until real transactions accrue.

C.2 Notifications — weekly overdue / at-risk milestone digest

What it does. A weekly digest of **overdue and at-risk** RAP milestones across the Index, delivered to an Institute inbox (in-app) and by email. "Overdue" = the target year has passed without confirmation; "at-risk" = due this year with low or stalled progress. There's a manual "Generate & send now" button for demos. It closes a gap where this risk signal was computed but never delivered.

How it works. A weekly schedule (and the manual button) compute the digest deterministically (reusing the same risk logic as the dashboards), **save the in-app record first** (so the inbox survives an email failure), then attempt email via AWS SES. Records are idempotent per ISO week — one row per week, re-runnable without duplicates.

> **Caveats.** - **Single recipient.** There is one sender and one recipient address — not a distribution list. No per-company digests, per-user preferences, or unsubscribe. - **Email is in the AWS "sandbox."** SES will only send to **verified** addresses (not arbitrary staff inboxes) until AWS production access is requested — which has **not** been done. Sender *and* recipient must be verified, per region. - **The weekly automated digest is production-only.** The scheduled weekly cron runs **only on `production`** — dev and `ca` stages deliberately have no weekly schedule (so they never emit stray emails); on `ca`, the digest is sent only via the manual "Generate & send now" button (which was verified end-to-end). On production it records **"skipped"** each week until the sender/recipient are configured (the in-app inbox still works); and since that config is baked in at deploy time, a deploy that forgets it **silently reverts to "skipped"** with no warning. - **Institute-only** — there are no company-facing notifications today (company demo logins are placeholders and undeliverable). - Digest content is aggregate, PII-free commitment data by design.

Where to go next

- Costs, risks, and the sandbox → client-owned account transition → 01 · Project Audit
- Standing the platform up in your own account → 02 · Deploy Runbook
- Which extraction engine to use, and CA validity → 03 · Engine Comparison
- Monitoring & security in plain language → 04 · Monitoring & Security Brief